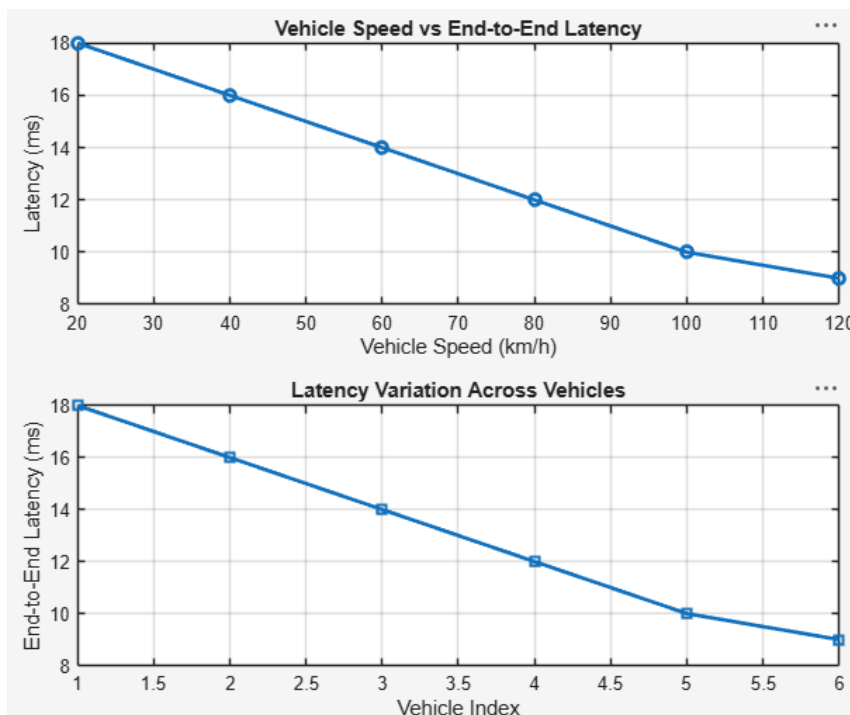


## Ex.No.13: V2X Latency Comparison

### Objective 1: Matlab Code and Output

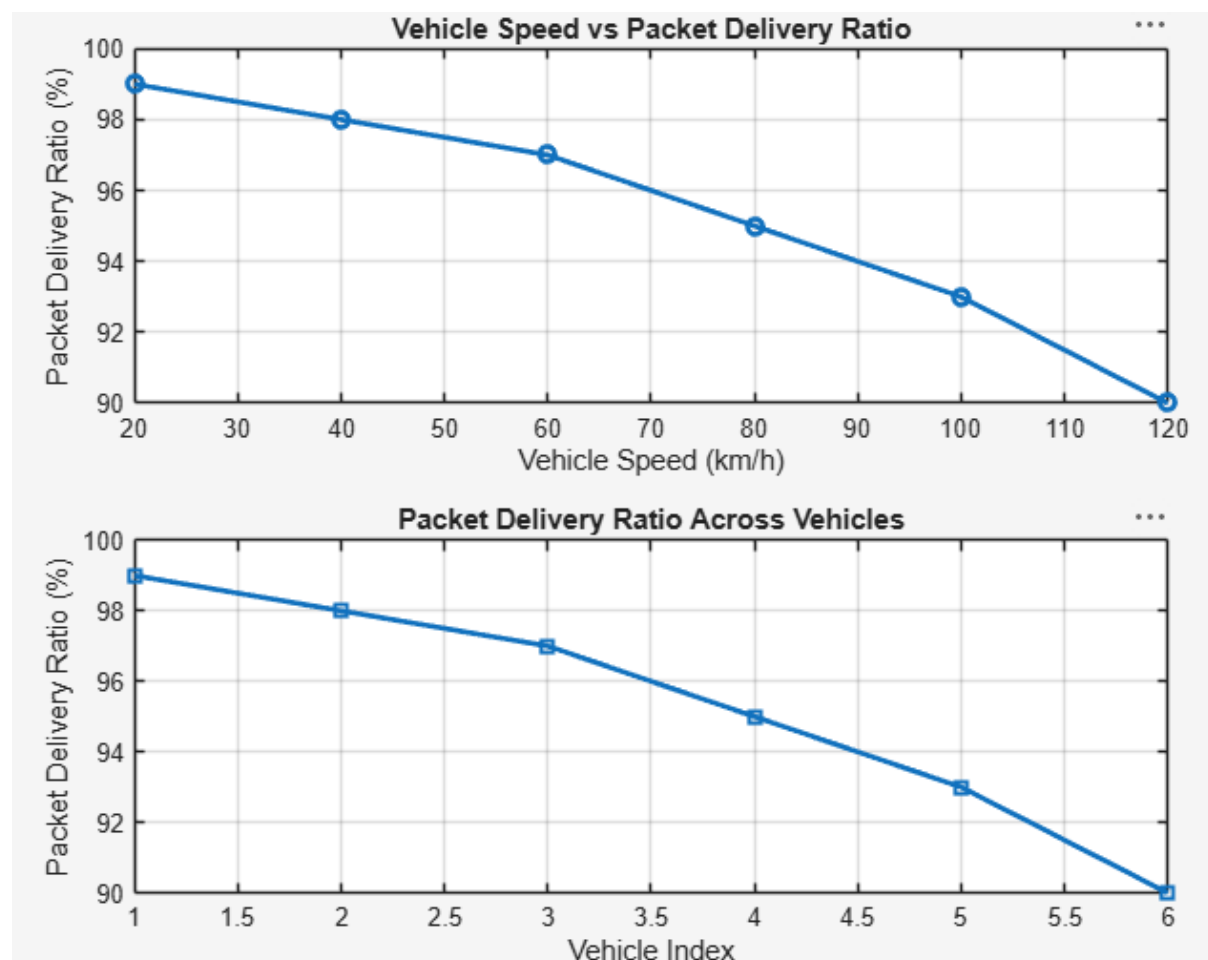
/MATLAB Drive/slotb13.m

```
1   clc;
2   clear;
3   close all;
4   % Vehicle Speed (km/h)
5   speed = 20:20:120;
6   % Simulated End-to-End Latency (ms)
7   latency = [18 16 14 12 10 9];
8   disp('Vehicle Speed (km/h)    End-to-End Latency (ms)')
9   disp([speed' latency'])
10  figure
11  % ----- Graph 1 -----
12  subplot(2,1,1)
13  plot(speed,latency,'-o','LineWidth',2)
14  title('Vehicle Speed vs End-to-End Latency')
15  xlabel('Vehicle Speed (km/h)')
16  ylabel('Latency (ms)')
17  grid on
18  % ----- Graph 2 -----
19  subplot(2,1,2)
20  plot(1:length(latency),latency,'-s','LineWidth',2)
21  title('Latency Variation Across Vehicles')
22  xlabel('Vehicle Index')
23  ylabel('End-to-End Latency (ms)')
24  grid on
```



## Objective 2: Matlab Code and Output

```
26 % Vehicle Speed (km/h)
27 speed = 20:20:120;
28 % Simulated Packet Delivery Ratio (%)
29 PDR = [99 98 97 95 93 90];
30 disp('Vehicle Speed (km/h)    Packet Delivery Ratio (%)')
31 disp([speed' PDR'])
32 figure
33 % ----- Graph 1 -----
34 subplot(2,1,1)
35 plot(speed,PDR,'-o','LineWidth',2)
36 title('Vehicle Speed vs Packet Delivery Ratio')
37 xlabel('Vehicle Speed (km/h)')
38 ylabel('Packet Delivery Ratio (%)')
39 grid on
40 % ----- Graph 2 -----
41 subplot(2,1,2)
42 plot(1:length(PDR),PDR,'-s','LineWidth',2)
43 title('Packet Delivery Ratio Across Vehicles')
44 xlabel('Vehicle Index')
45 ylabel('Packet Delivery Ratio (%)')
46 grid on
```



### Objective 3: Matlab Code and Output

```
48 % Vehicle IDs
49 vehicle = 1:6;
50 % Vehicle Speed (km/h)
51 speed = [30 45 60 75 90 105];
52 % Communication Range (m)
53 range = [280 260 240 220 200 180];
54 disp('Vehicle Speed(km/h) Communication Range(m)')
55 disp([vehicle' speed' range'])
56 figure
57 % ----- Graph 1 -----
58 subplot(2,1,1)
59 plot(vehicle,speed,'-o','LineWidth',2)
60 title('Vehicle Speed of Different Vehicles')
61 xlabel('Vehicle Index')
62 ylabel('Vehicle Speed (km/h)')
63 grid on
64 % ----- Graph 2 -----
65 subplot(2,1,2)
66 plot(speed,range,'-s','LineWidth',2)
67 title('Vehicle Speed vs Communication Range')
68 xlabel('Vehicle Speed (km/h)')
69 ylabel('Communication Range (m)')
70 grid on
```

